Aedin Lai

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EDUCATION

Northeastern University, Boston, MA *Expected May 2029*
Candidate for Bachelor of Science in Electrical & Computer Engineering
Relevant Coursework: Embedded Design, Computing Fundamentals for Engineers, Physics 2, Calculus 2

EXPERIENCE

Data Center Infrastructure Intern | SingleStore *June – September 2026*

Boston, MA

- Led hardware deployment for a phased data center expansion, scaling a private cloud platform from 160 to 300+ VMs
- Replaced a single-push rollout with a phased "two racks at a time" plan, bringing three cohorts of 40–50 VMs online while limiting disruption to live workloads
- Diagnosed a Linux kernel storage defect and a multi-week NFS instability, restoring full fleet capacity
- Built a Go audit tool and REST API that checks power supply, memory, NIC, and drive health across the server fleet via iDRAC
- Supported a full Oakland-to-Boston data center relocation, migrating the platform from bare-metal to virtualized infrastructure

Research Intern | PEACE Foundation, Yale University School of Medicine *2023 – 2024*

New Haven, CT — Supervisor: Dr. Kun Liu, Associate Professor of Sports Neuroscience

- Designed, built, and improved a full-body balance measurement device to study cognition and movement in neurological patients
- Automated data collection using custom hardware across ASD, Alzheimer's, Parkinson's, and schizophrenia patient populations
- Presented at an international conference (ISEC 2024); co-authored a study accepted to *JAMA* (2025)

PROJECTS

aedinlai.com | Personal Site & Project Hub *February 2025 – Present*

JavaScript, Cloudflare Workers, SQL, GitHub Actions

- Built a framework-free single-page site hosting my live projects and a daily auto-generated news feed
- Engineered a gym occupancy tracker that records live counts every 10 minutes into a serverless SQL database
- Designed a serverless SQL API for a shared pizza-pricing app, with per-IP rate limiting and owner-only edits

Sandbit | Pixel Physics Engine *September 2026*

Rust, WebAssembly, WebGL

- Built an 5,700-line Rust/WebAssembly falling-sand engine simulating 27 materials, including electrical conduction
- Implemented rigid-body physics with contact manifolds, buoyancy, and rope and motor joints for pixel objects
- Verified with 30 physics tests, fuzzing, and a headless player that runs all 15 levels across 5 random seeds

One-Inch Punch Arcade Machine | Cornerstone of Engineering *September – December 2025*

Laser Sensors, Electronics, Mechanical Design

- Designed a dual-laser-sensor trigger and wrote scoring logic that converts swing velocity into a live score
- Led the full build, from electronics to a pendulum-and-magnet auto-reset, and validated it with 80+ full-force stress tests
- Identified a paddle material limit during a full day of expo play and proposed a rubber-composite redesign

Hallway Marathon | Event Website *November 2025*

HTML, CSS, JavaScript

- Built a framework-free event site with live betting odds and competitor stats, run live for a multi-university race

More projects at aedinlai.com and github.com/Aedin-ctrl

LEADERSHIP & ATHLETICS

Captain | Northeastern University Club Fencing *2025 – Present*

- Recruit new fencers and design practices; anchor the epee squad at the 2026 Collegiate Club National Championship

Competitive Epee Fencer | Marx Fencing Academy *2018 – 2025*

- Nationally ranked #103 in the U.S.; Top-16 World Cup finish; All-American 2021–2025

SKILLS

Programming: Python • Java • C# • Rust • SQL • MATLAB • JavaScript • HTML/CSS
Hardware & CAD: Quartus Prime • Raspberry Pi • Sensor Integration • SolidWorks • Fusion 360 • AutoCAD • Laser Cutting
Tools: Git • GitHub Actions • Linux • Cloudflare Workers • WebAssembly • VS Code